
VALOVIR (Valaciclovir Tablet 500mg)

DESCRIPTION AND COMPOSITION

White to off-white, film coated, caplet shaped tablets, debossed with 'V' on one side and '500' on other side. Each film coated tablet contains: Valaciclovir Hydrochloride 556mg equivalent to Valaciclovir 500mg.

PHARMACODYNAMIC PROPERTIES

Valaciclovir an antiviral, is the L-valine ester of acyclovir. Aciclovir is a purine (guanine) nucleoside analogue.

Mode of action:

Mode of action: Valaciclovir is rapidly and almost completely converted in man to acyclovir and valine, probably by the enzyme referred to as valaciclovir hydrolase.

Aciclovir is a specific inhibitor of the herpes viruses with in vitro activity against herpes simplex viruses (HSV) type 1 and type 2, varicella zoster virus (VZV), cytomegalovirus (CMV), Epstein Bair Virus (EBV) and human herpes virus 6 (HHV-6). Aciclovir inhibits herpes virus DNA synthesis once it has been phosphorylated to the active triphosphate form. The first stage of phosphorylation requires the activity of a virus-specific enzyme. In the case of HSV, VZV and EBV this enzyme is the viral thymidine kinase (TK), which is only present in virus infected cells. Selectivity is maintained in CMV with phosphorylation, at least in part, being mediated through the phosphotransferase gene product of UL97.

This requirement for activation of acyclovir by a virus specific enzyme largely explains its selectivity. The phosphorylation process is completed conversion from mono to triphosphate by cellular kinases. Aciclovir triphosphate competitively inhibits the virus DNA polymerase and incorporation of this nucleoside analogue results in obligate chain termination, halting virus DNA synthesis and this blocking virus replication. Extensive monitoring of clinical HSV and VZV isolates from patients receiving acyclovir therapy or prophylaxis has revealed that virus with reduced sensitivity to acyclovir is extremely rare in the immunocompetent and is only found infrequently in severely immuno compromised individuals e.g. organ or bone marrow transplant recipients, patients receiving chemotherapy for malignant disease and people infected with the human immunodeficiency virus (HIV).

Resistance is normally due to thymidine kinase deficient phenotype which results in a virus which is profoundly disadvantage in the natural host. Infrequently, reduced sensitivity to acyclovir has been described as a result of subtle alteration in either the virus thymidine kinase or DNA polymerase. The virulence of these variants resembles that of the wild type of virus.

PHARMACOKINETIC PROPERTIES

General characteristics:

After oral administration valaciclovir is well absorbed and rapidly and almost completely converted to acyclovir and valine. This conversion is probably mediated by an enzyme isolated from human liver referred to as valaciclovir hydrolase. In patients with normal renal function the plasma elimination half-life of aciclovir after both single and multiple dosing with valaciclovir is approximately 3 hours. In patients with end-stage renal disease, the average elimination half-life of aciclovir administration is approximately 14 hours. Less than 1% of the administered dose of valaciclovir is recovered in the urine as unchanged drug. Valaciclovir is eliminated in the urine principally as aciclovir (greater than 80% of the recovered dose) and the known aciclovir metabolite, 9 (carboxymethoxymethylguanine (CMMG)).

Characteristics in patients: Herpes zoster and herpes simplex do not significantly alter the pharmacokinetics of valaciclovir and aciclovir after oral administration of Valaciclovir.

THERAPEUTIC INDICATION:

Valaciclovir is indicated for the treatment of herpes zoster (shingles).

Valaciclovir accelerates the resolution of pain. It reduces the duration of and the proportion of patients with zoster associated pain, which includes acute and post-herpetic neuralgia.

Valaciclovir can prevent lesion development when taken at the first signs and symptoms of an HSV recurrence.

Valaciclovir is indicated for the prevention (suppression) of recurrent herpes simplex infections of the skin and mucous membranes, including genital herpes.

Valaciclovir can reduce transmission of genital herpes when taken as suppressive therapy and combined with safer sex practices (particularly the use of condoms).

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DOSAGE:

Treatment of herpes zoster:

The dosage in adults is 1000mg of Valaciclovir to be taken 3 times daily for 7 days.

Treatment of herpes simplex infections:

The dosage in adults is 500 mg of Valaciclovir to be taken twice daily. For recurrent episodes, treatment should be for 5 days.

For initial episodes, which can be more severe, treatment may have to be extended to 10 days. Dosing should begin as early as possible. For recurrent episodes of herpes simplex, this should ideally be during the prodromal period or immediately the first signs or symptoms appear.

Prevention (suppression) of recurrence of herpes simplex infections:

In immunocompetent adult patients, 500 mg of Valaciclovir is to be taken once daily. Some patients with very frequent recurrences (eg 10 or more per year) may gain additional benefit from the daily dose of 500mg being taken as a divided dose (250mg twice daily). For immunocompromised adult patients the dose is 500mg twice daily.

Reduction of transmission of genital herpes:

In immunocompetent heterosexual adult patients with 9 or fewer recurrence per year; 500 mg Valaciclovir to be taken once daily by the infected partner in order to reduce transmission to a sexual partner negative for HSV-2 antibodies. Safer sex practices (particularly condom use) should be maintained, and sexual contact avoided if lesions are present.

There are no data on the reduction of transmission beyond 8 months in other patient populations.

Dosage in renal impairment:

Herpes zoster treatment and herpes simplex treatment and prevention (suppression): The dosage of Valaciclovir should be reduced in patients with significantly impaired renal function as shown in the table below:

Therapeutic Indication	Creatinine clearance ml/min	Valaciclovir dosage
Herpes zoster	15- 30	1g twice a day
	Less than 15	1g once a day
Herpes simplex (treatment) (500mg bd regimen)	Less than 15	500mg once a day
<i>Herpes simplex prevention:</i>		
<i>-immunocom- petaint patients</i>	Less than 15	250mg once a day
<i>-immunocom- promised patients</i>	Less than 15	500mg once a day
Reduction of transmission of genital Herpes	Less than 15	250mg once a day

In patients on haemodialysis, the Valaciclovir dosage recommended for patients with a creatinine clearance of less than 15ml/min should be used. This should be administered after the hemodialysis has been performed.

Dosage in hepatic impairment: 1g unit dose of Valaciclovir show that modification is not required in patients with mild or moderate cirrhosis (hepatic synthetic function maintained). Pharmacokinetic data in patients with advanced cirrhosis, (impaired hepatic synthetic function and evidence of portal-systemic suffering) do not indicate the need for dosage adjustment; however, clinical experience is limited.

For higher doses (4g or more/day).

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Warning and special precautions for use:

Dosage of children:

There are no data available on the use of Valaciclovir in children.

Dosage in the elderly:

Dosage modification is not required unless renal function is significantly impaired.

Adequate hydration should be maintained.

CONTRAINDICATIONS

Valaciclovir is contra-indicated in patients known to be hypersensitive to valaciclovir, aciclovir or any components of formulations of Valaciclovir.

SPECIAL WARNING AND SPECIAL PRECAUTIONS FOR USE

Hydration status:

Care should be taken to ensure adequate fluid intake in patients who are at risk of dehydration, particularly the elderly.

Use in renal impairment:

The Valaciclovir dose should be adjusted in patients with renal impairment. Patients with a history of renal impairment are also at increased risk of developing neurological effects.

Use of higher doses of Valaciclovir in hepatic impairment and liver transplantation:

These are no data available on the use of higher dose of Valaciclovir (4g or more/day) in patients with liver disease.

Caution should therefore be exercised when administering higher doses of Valaciclovir to these patients.

Use in genital herpes:

Suppressive therapy with Valaciclovir reduces the risk of transmitting genital herpes. It does not cure genital herpes or completely eliminate the risk of transmission. In addition to therapy with Valaciclovir it is recommended that patients use safer sex practices (particularly the use of condoms).

Immune:

Drug reaction with eosinophilia and systemic symptoms (DRESS), which can be life-threatening or fatal, has been reported in association with valaciclovir treatment. Patients should be advised of the signs and symptoms and monitored closely for skin reactions. If signs and symptoms suggestive of DRESS appear, valaciclovir should be withdrawn immediately and an alternative treatment considered (as appropriate). If a patient has developed DRESS with the use of valaciclovir, treatment with valaciclovir must not be restarted in this patient at any time.

INTERACTION WITH OTHER MEDICINAL PRODUCTS AND OTHER FORMS OF INTERACTIONS

No clinically significant interactions have been identified. Following 1g Valaciclovir, cimetidine and probenacid increase the AUC of aciclovir by this mechanism and reduce aciclovir renal clearance. However, no dosage adjustment is necessary at this dose because of the wide therapeutic index of aciclovir. In patients receiving higher doses of Valaciclovir caution is required during concurrent administration with drugs which compete with aciclovir for elimination, because of the potential for increased plasma levels of one or both drugs or their metabolites. Increases in plasma AUC of aciclovir and of the inactive metabolite of mycophenolate mofetil, an immuno suppressant agent used in transplant patients; have been shown when the drugs are co administered. Care is also required (with monitoring for changes in renal function) if administering higher dose of Valaciclovir (4g or more/day) with drugs which affect other aspects of renal physiology (eg. Cyclosporine, tacrolimus).

PREGNANCY AND LACTATION

Teratogenicity:

Valaciclovir was not teratogenic in rats or rabbits. Valaciclovir is almost completely metabolized to aciclovir. Subcutaneous administration of aciclovir in internationally accepted tests did not produce teratogenic effects in rats or rabbits. In additional studies in rats, foetal abnormalities were observed at subcutaneous doses that produced plasma levels of 100 mcg/ml and maternal toxicity.

Fertility:

Valaciclovir did not affect fertility in male or female rats dosed by the oral route.

Pregnancy:

There are limited data on the use of Valaciclovir in pregnancy. Valaciclovir should only be used in pregnancy if the potential benefits of treatment outweigh the potential risk. Pregnancy registries have documented the pregnancy outcomes in women exposed to Valaciclovir or to any formulation of aciclovir. The findings of the aciclovir pregnancy

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registry have not shown an increase in the number of birth defects amongst aciclovir-exposed subjects compared with the general population, and any birth defects showed no uniqueness or consistent pattern to suggest common cause.

Lactation:

Aciclovir, the principle metabolite of Valaciclovir, is excreted in breast milk. Unchanged Valaciclovir was not detected in maternal serum, breast milk, or infant urine. Caution is advised if Valaciclovir is to be administered to a nursing woman. However, Aciclovir is used to treat neonatal herpes simplex at intravenous dose of 30 mg/kg/day.

SIDE EFFECTS

Adverse reactions are listed below by MedDRA body system organ class and by frequency.

The frequency categories used are:

Very common ≥ 1 in 10,

Common ≥ 1 in 100 and < 1 in 10,

Uncommon ≥ 1 in 1,000 and < 1 in 100,

Rare ≥ 1 in 10,000 and < 1 in 1,000,

Very rare < 1 in 10,000.

Very rare < 1 in 10,000.

Blood and lymphatic system disorders

Very rare: Leukopenia / Neutropenia, thrombocytopenia. Leukopenia / Neutropenia is mainly reported in immuno compromised patients.

Immune system disorders

Very rare: Anaphylaxis.

Drug reaction with eosinophilia and systemic symptoms (DRESS).

Psychiatric and nervous system disorders

Rare: Dizziness, confusion, hallucinations, decreased consciousness.

Very rare: Agitation, tremor, ataxia, dysarthria, psychotic symptoms convulsions, encephalopathy, coma. The above events are generally reversible and usually seen in patients with renal impairment or with other predisposing factors.

Respiratory, thoracic and mediastinal disorders

Uncommon: Dyspnoea. Gastrointestinal disorders

Common: *Nausea

Rare: Abdominal discomfort, vomiting, diarrhoea.

Hepato-biliary disorders

Very rare: Reversible increases in liver function tests. These are occasionally described as hepatitis.

Skin and subcutaneous tissue disorders

Uncommon: Rashes including photosensitivity.

Rare: Pruritus.

Very rare: Urticaria, angioedema.

Renal and urinary disorders

Rare: Renal impairment.

Very rare: Acute renal failure, renal pain.

Renal pain may be associated with renal failure.

Other: There have been reports of renal insufficiency, microangiopathic haemolytic anaemia and thrombocytopenia (sometimes in combination) in severely immunocompromised patients, particularly those with advanced HIV disease, receiving high doses (8 g daily) of valaciclovir for prolonged periods in clinical trials. These findings have been observed in patients not treated with valaciclovir who have the same underlying or concurrent conditions.

SYMPTOMS AND TREATMENT OF OVERDOSE

Symptoms and signs: There are at present limited data available on overdosage with Valaciclovir. However, patients have ingested single overdoses of up to 20g of aciclovir, which is only partially absorbed in the gastrointestinal tract, usually without toxic effects. Accidental, repeated overdoses of oral aciclovir over several days have been associated with gastrointestinal effects (such as nausea and vomiting) and neurological effects (headache and confusion). Overdose of intravenous aciclovir has resulted in elevations of serum creatinine and subsequent renal failure. Neurological effects including confusion, hallucinations, agitation, seizures and coma have been described in association with intravenous overdosage.

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Management: Patients should be observed closely for signs of toxicity. Hemodialysis significantly enhances the removal of aciclovir from the blood and may, therefore, be considered a management option in the event of symptomatic overdose.

STORAGE

Store below 30°C. Protect from light.

PRESENTATION

Blister strip of 10 tablets, box of 10 x 10's.

Manufactured by:

HETERO LABS LIMITED

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Product Registration Holder:

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